

Executive Business Report

Superstore Sales Forecasting & Demand Intelligence System

Executive Summary

This report presents an analysis of four years of Superstore sales data to understand sales trends, forecast future demand, identify unusual sales patterns, and segment products based on demand characteristics. The project combines statistical forecasting, machine learning, anomaly detection, and clustering techniques to support inventory planning and financial decision-making. The analysis shows strong seasonal sales patterns, identifies the Technology category as the highest revenue generator, and highlights opportunities for improving stock planning and operational efficiency.

Key Findings from Data Exploration and Forecasting

Sales Performance

Analysis of the Superstore dataset revealed several important business insights:

- **Technology** generated the highest total revenue with approximately **\$827,456**, followed by Furniture and Office Supplies.
- The **East** region demonstrated the most consistent year-over-year sales growth throughout the four-year period.
- The average time between order placement and shipment was **3.96 days**, with only minor differences across regions.
- Sales exhibit strong seasonality, with the highest sales consistently occurring during **September, November, and December**, indicating increased customer demand during festive and year-end shopping periods.

The time series decomposition further confirmed an upward long-term sales trend with recurring seasonal peaks.

Three-Month Sales Forecast

Three forecasting approaches were evaluated:

- SARIMA
- Facebook Prophet
- XGBoost Regressor

The selected forecasting model predicts that overall sales will continue following the historical upward trend while maintaining the observed seasonal behaviour. Based on historical performance, sales are expected to remain within the confidence intervals produced by the forecasting model under normal business conditions.

The three-month forecast provides management with an opportunity to prepare inventory procurement, warehouse capacity, staffing, and financial budgets in advance. Higher inventory levels should be maintained before the expected seasonal demand peaks to reduce stock shortages.

Anomaly Detection

Two independent anomaly detection techniques were applied:

- Isolation Forest
- Z-Score Detection

Both methods successfully identified unusual sales weeks that differed significantly from normal purchasing behaviour.

Major Observations

1. Sales spikes during **November** are likely associated with festive shopping seasons, promotional campaigns, and holiday discounts.
2. Higher-than-average sales during **December** indicate increased year-end consumer spending and holiday purchasing.
3. A few unusually low sales weeks may correspond to temporary supply chain issues, reduced customer activity, or seasonal demand fluctuations.

Using both anomaly detection techniques together improves reliability because machine learning identifies unusual behaviour while statistical methods validate significant deviations from historical averages.

Product Demand Segmentation

Product sub-categories were segmented using K-Means clustering based on:

- Total Sales
- Sales Growth Rate
- Sales Volatility
- Average Order Value

Four meaningful demand groups were identified.

Recommended Stocking Strategy

High Volume, Stable Demand

- Maintain high inventory levels.
- Prioritise continuous replenishment.
- Ensure products remain consistently available.

Growing Demand

- Increase inventory gradually.
- Monitor demand frequently.
- Expand supplier capacity where necessary.

Low Volume, High Volatility

- Keep limited stock.
- Use demand-driven purchasing.
- Avoid excess inventory carrying costs.

Declining Demand

- Reduce purchasing quantities.
- Focus on promotional sales.
- Minimise warehouse space allocated to these products.

Business Recommendations

1. Increase Investment in Technology Products

Technology contributes the highest overall revenue (approximately **\$827,456**). Increasing inventory availability and marketing investment for this category can maximise future revenue while reducing stock-out risk.

2. Prepare Inventory Before Seasonal Peaks

Historical sales consistently increase during **September through December**, particularly in November and December. Inventory procurement and warehouse planning should be completed before these months to satisfy increased customer demand.

3. Focus Growth Strategies in the East Region

The East region demonstrates the most consistent long-term sales growth across the dataset. Additional promotional campaigns, distribution capacity, and inventory allocation in this region are likely to generate stronger returns than slower-growing regions.

Risk and Limitation

The forecasting models rely entirely on historical sales data. Unexpected events such as economic downturns, changes in customer behaviour, supplier disruptions, inflation, or new competitors may reduce forecast accuracy. The forecasting model should therefore be retrained regularly using the latest sales data to maintain prediction performance.

Conclusion

This project demonstrates how data analytics can improve retail decision-making through sales forecasting, anomaly detection, and demand segmentation. The insights generated help optimise inventory management, improve financial planning, reduce operational costs, and support better supply chain decisions. Regular model updates and continuous monitoring will enable the organisation to respond effectively to changing market conditions and sustain long-term business growth.